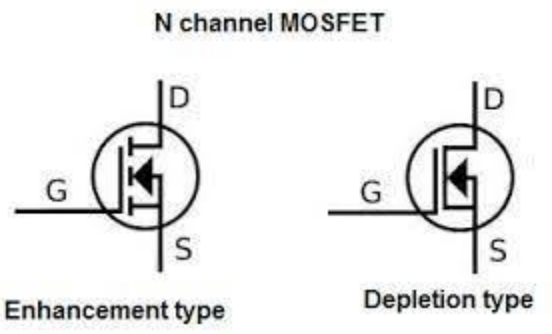
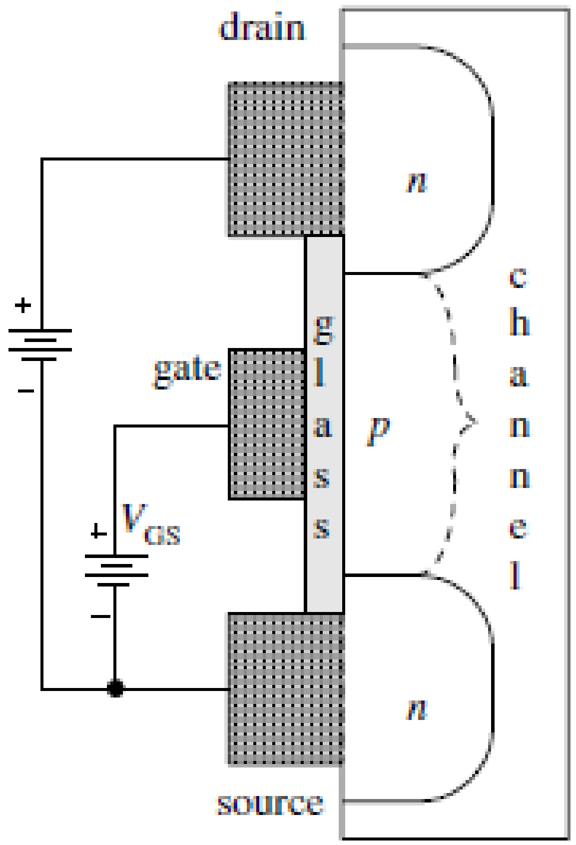
POWER ELECTRONICS

Enhancement Mode MOSFET

Enhancement Mode MOSFET

- N-channel enhancement mode Metal Oxide Semiconductor Field Effect Transistor
- Three connections: source, drain, gate
- The channel with p-type material (holes) (open circuit)
- Glass (silicon dioxide), insulator
- 0 volts at gate, no current in transistor, off



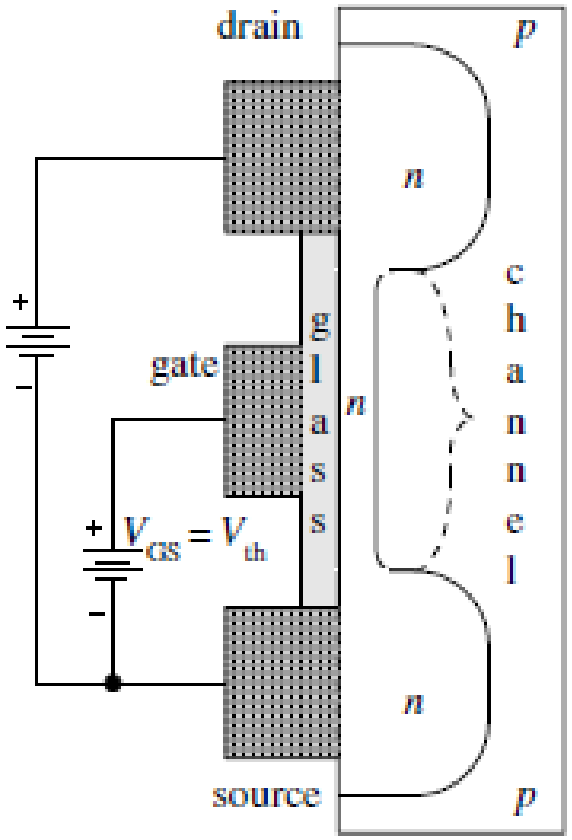


Enhancement Mode MOSFET

- Positive voltage at gate, conduction
- N-channel path between drain and source
- More positive voltage, more conduction
- Gate-to-source voltage needed to enhance the channel to allow 250 μA of drain (or source) current is called the threshold voltage

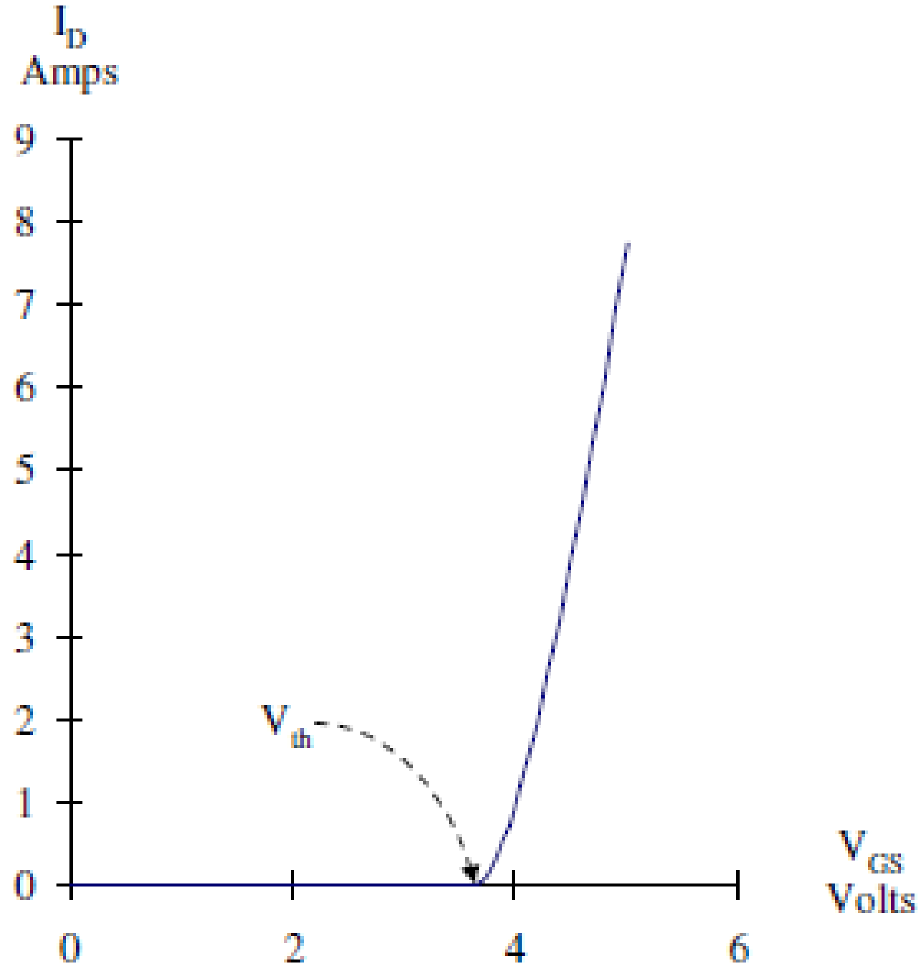
$$V_{th} = V_{GS}\Big|_{I_D=250\mu A}$$

- Turning on MOSFET, apply voltage more than V_{th}
- Beyond threshold voltage, drain current is proportional to square of the Gate-to-source voltage



Enhancement Mode MOSFET

- I_D versus V_{GS} plot
- IRF530N
- Once transistor is on, drain current is relatively independent of drain-to-source voltage



Enhancement Mode MOSFET

- You got knowledge about the n-channel MOSFET, what about p-channel?
- What are the changes?
- Draw the circuit and identify the changes keeping in mind the configuration of n-channel.

Enhancement Mode MOSFET

- As temperature rises, drain current increases
- Dissipating more power
- Heating up the transistor more,
- Producing more current,
- Heating up the transistor more, so on and so on
- MOSFET may thermally run away, burn themselves up
- Proper heat sinks are required in op-amps using MOSFETs

Heat Sinks

- The hotter the IC gets, the poorer its performance and sooner it will fail
- Critical to remove heat from silicon
- Heat sinks is central to success of any power electronic device

$$T_J = T_A + P\left(\Theta_{JC} + \Theta_{CS} + \Theta_{SA}\right)$$

$T_J =$ Silicon junction temperature (specification)

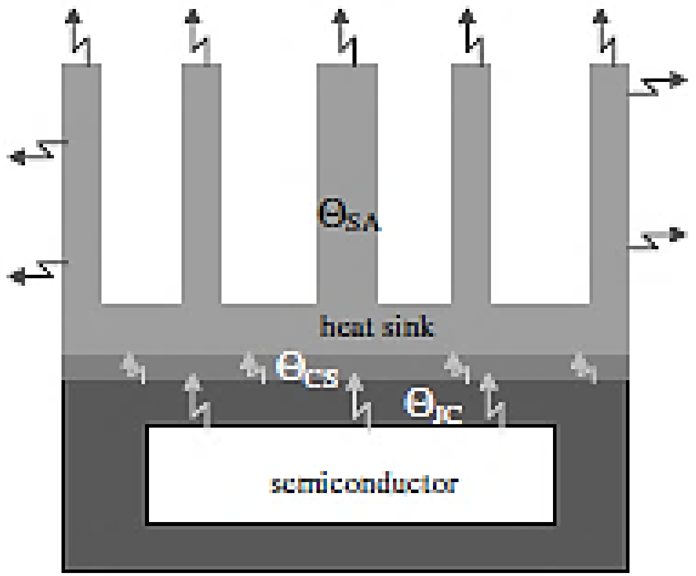
$T_A =$ Ambient temperature (surrounding air)

$P =$ Power that IC must dissipate

$\Theta_{JC} =$ thermal resistance between junction and case (specification)

$\Theta_{CS} =$ thermal resistance between case and heat sink

$\Theta_{SA} =$ thermal resistance between heat sink and surrounding air



Heat Sinks

- Cooler the silicon, longer the IC will work
- Keep $T_J \leq 125°\text{C}$

Example

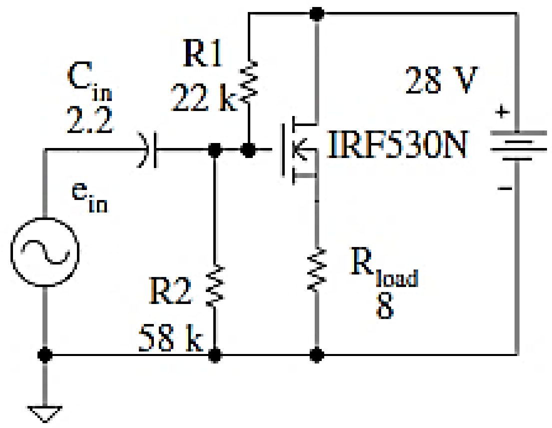
- If a heat sink with a thermal resistance of $4\,\frac{°\mathrm{C}}{\mathrm{W}}$ is used with $\pm$ 24 V supplies, what is the largest sine wave that can be delivered to a 10 Ω load? Consider the ambient temperature of 40 degrees and the maximum IC temperature is 125 degree centigrade. Also consider $\Theta_{JC} = 2.5$°C and $\Theta_{CS} = 2$°C

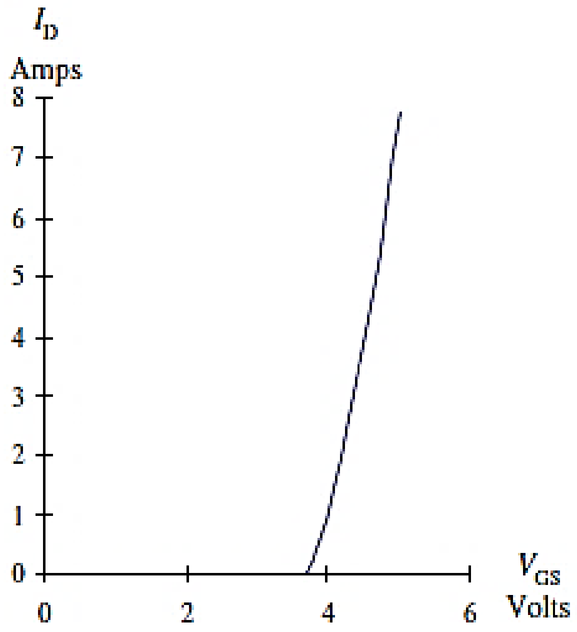
CLASS A

Common Drain Amplifier

Common Drain Amplifier

- Class A amplifier conducts during the entire cycle of input signal
- Input reproduced at the output
- Lowest distortion
- Why it is called Common Drain?





DC Operation

- Voltage divider
- Gate Voltage

AC Operation

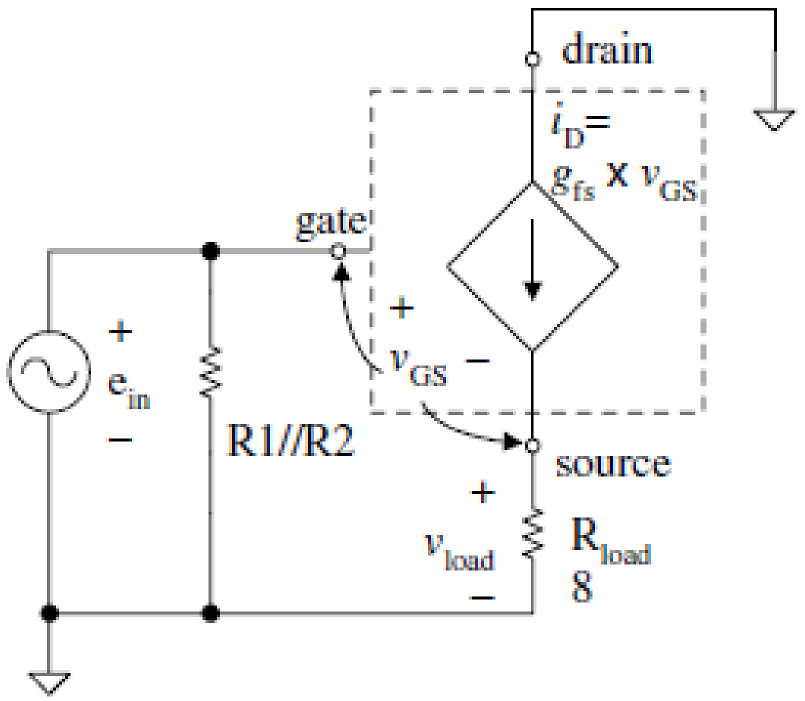
- Amplifier’s forward transconductance

$$g_{fs} = \left.\frac{\Delta I_D}{\Delta V_{GS}}\right|_{V_{DS}\text{ constant}}$$

- For IRF530 it is greater than 6.4 A/V
- Apply KVL for load voltage

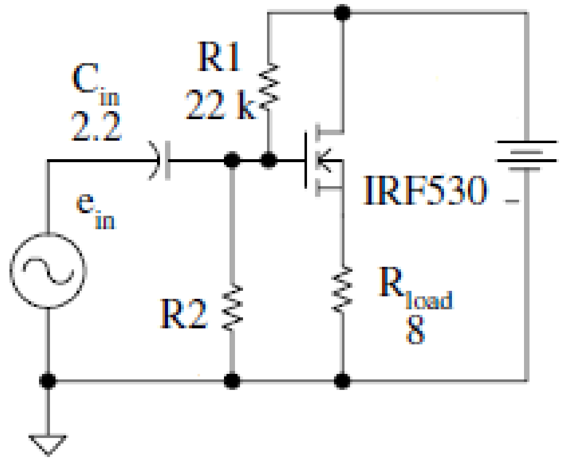
$$v_{load} = \frac{g_{fs}R_{load}}{1 + g_{fs}R_{load}} e_{in}$$

- Since the denominator is always more than numerator
- Load voltage is always less than the supply voltage
- $A_v < 1$
- Can supply several amps of current to load



Example

- Calculate the dc and rms voltages at gate, load and the dc and rms load current. Consider the $R_2 = 30\ \text{k}\Omega$, $V_{dc} = 56$ V and $e_{in} = 20$ V peak.



Problems with Class A Amplifier

- Load is directly connected to source terminal
- Bias current flows continuously even when no signal is applied

$$P_{supply} = 56 \times 3.5 = 196 \text{ W}$$

$$P_{load} = (3.5)^2 \times 8 = 98 \text{ W}$$

- Power dissipation in the transistor

$$P_Q = 196 - 98 = 98 \text{ W}$$

- This power is wasted. It is not produced by the signal. The output (loudspeaker) will burn out even before making any sound
- Class A amplifier is extremely wasteful.
- Rarely used

Reference

Power Electronics, Principles and Applications
J. Michael Jacob

Power Electronics
Ned Mohan